INCH-POUND

MIL-S-19500/584
25 July 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, RF POWER TRANSISTOR, NPN, SILICON, PUSH-PULL
TYPES JANTX, JANTXV, AND JANS, 2N6985 AND 2N6986

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a NPN, silicon, push-pull RF power transistor. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

| $P_T$ | | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $T_{stg}$ | $T_J$ |
|---|---|---|---|---|---|---|---|
| $T_C$ = 25°C 1/ | $T_C$ = 25°C 2/ | | | | | | |
| W | W | V dc | V dc | V dc | A dc | °C | °C |
| 135 | 270 | 60 | 30 | 4 | 16 | -65 to +200 | +200 |

1/ Derate linearly 0.77 W/°C for $T_C$ > 25°C (one section).
2/ Derate linearly 1.54 W/°C for $T_C$ > 25°C (both sections).

1.4 Primary electrical characteristics.

| $V_{CE(sat)}$ 1/ 2/ | | $C_{obo}$ | $h_{FE}$ | $G_{PE}$ 3/ | $G_{PE}$ 3/ |
|---|---|---|---|---|---|
| Limits | $I_C$ = 5 A dc<br>$I_B$ = 2 A dc | $I_E$ = 0<br>$V_{CB}$ = 28 V<br>f = 1 MHz | $V_{CE}$ = 5 V 1/<br>$I_C$ = 1 A | $V_{CC}$ = 28 V<br>f = 400 MHz<br>$P_{out}$ = 125 W | $V_{CC}$ = 28 V<br>f = 500 MHz<br>$P_{out}$ = 100 W |
| | V dc | pF | | 2N6985<br>dB | 2N6986<br>dB |
| Min | --- | --- | 20 | 8.0 | 7.5 |
| Max | 2.2 | 115 | 100 | --- | --- |

1/ Pulsed (see 4.5.1).
2/ Each side.
3/ See 6.2 for typical power output versus frequency.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronic Supply Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order.

3.4 Marking. Marking shall be in accordance with MIL-S-19500 except at the option of the manufacturer, the following marking may be omitted from the body of the transistor:

a. Country of origin.

b. Manufacturer's identification.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein (see 4.3.2 herein).

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II) of MIL-S-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 1/ | See 4.3.2 | See 4.3.2 |
| 9 | $I_{CBO}$, $h_{FE}$ | $I_{CBO}$, $h_{FE}$ |
| 11 | $\Delta I_{CBO}$ = 100% of initial value or 150 µA, whichever is greater; $\Delta h_{FE}$ = +25%; Subgroup 2 of table I herein. | $\Delta I_{CBO}$ = 100% of initial value or 150 µA, whichever is greater; $\Delta h_{FE}$ = ±25%; Subgroup 2 of table I herein. |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | $\Delta I_{CBO}$ = 100% of initial value or 150 µA, whichever is greater; $\Delta h_{FE}$ = ±25%; Subgroup 2 of table I herein. | $\Delta I_{CBO}$ = 150% of initial value or 100 µA, whichever is greater; $\Delta h_{FE}$ = ±25%; Subgroup 2 of table I herein. |

1/ Shall be performed anytime before screen 10.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$V_{CB} \geq 10$ V; $R_{\theta JC}$ = +0.65°C/W (both sections).
$T_J$ = +187.5°C ±12.5°C (see 4.5.3).
$T_A$ = +30°C ±5°C.

4.3.2 Thermal response ($\Delta V_{BE}$ measurements). The $\Delta V_{BE}$ measurements shall be performed using Tesec $\Delta V_{BE}$ tester, model number 8114/8115 or equivalent. The following measurement conditions shall apply. The delta $V_{BE}$ conditions and maximum $V_{BE}$ limit shall be derived by each vendor and the read and record mesurements shall be submitted in the qualification report. Read and record all $\Delta V_{BE}$ measurements and conditions for each device in the qualification lot. In addition, the qualification report shall include a thermal response curve. The chosen $\Delta V_{BE}$ values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots.

a. All test conditions ±5%.

b. $V_{CE}$ = 20 V.

c. $I_C$ = 5 amps.

d. Heating pulsewidth = 400 ms.



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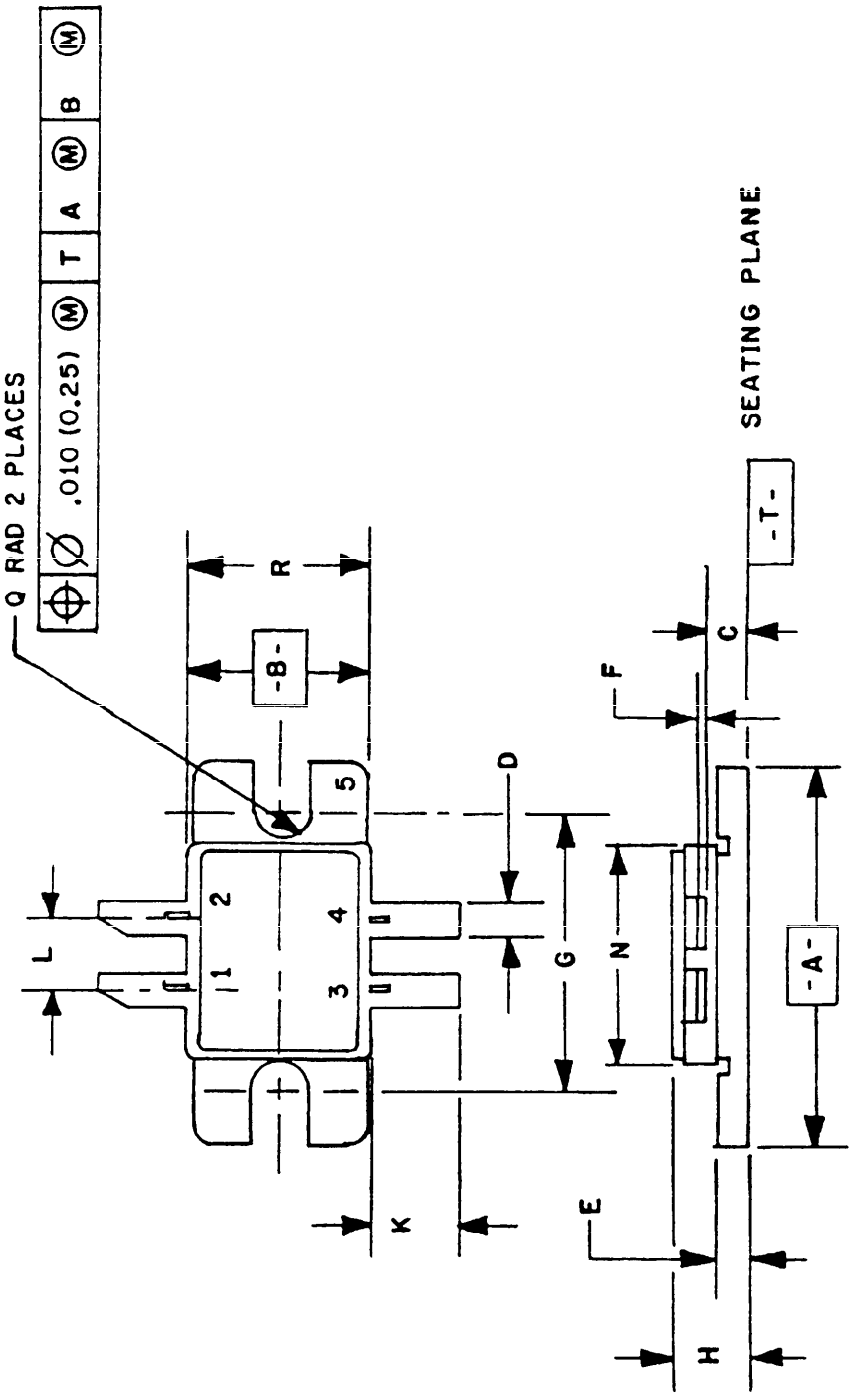
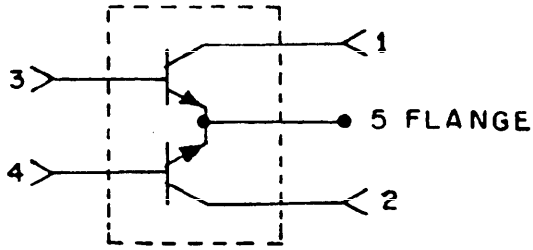



FIGURE 1. Dimensions and configuration.



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STYLE 1

STYLE 1:
PIN 1. COLLECTOR
2. COLLECTOR
3. BASE
4. BASE
5. EMITTER

SCHEMATIC DIAGRAM

| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .890 | .910 | 22.61 | 23.11 |
| B | .370 | .400 | 9.40 | 10.16 |
| C | .105 | .135 | 2.67 | 3.43 |
| D | .065 | .085 | 1.65 | 2.16 |
| E | .055 | .065 | 1.40 | 1.65 |
| F | .003 | .006 | 0.08 | 0.15 |
| G | .650 | BSC | 16.51 | BSC |
| H | .150 | .175 | 3.81 | 4.44 |
| K | .190 | .210 | 4.83 | 5.33 |
| L | .155 | .190 | 3.94 | 4.83 |
| N | .490 | .510 | 12.45 | 12.95 |
| Q | .060 | .070 | 1.52 | 1.78 |
| R | .390 | .410 | 9.91 | 10.41 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration - Continued.



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e. Measurement delay = 25 μs (after heating pulse).

f. Measurement current $V_{BE}$ = 10 mA.

g. Test limits: $V_{BE}$ (each side = 400 mV max. -side-to-side (1) $V_{BE}$ = ±20%.
(1) Δ imbalance = (x-y/x) *100 Where x = larger $V_{BE}$ reading
Y = smaller $V_{BE}$ reading

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein).

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in MIL-STD-750.

4.5.2 Testing of units. All specified electrical tests, including end-points tests, shall be performed equally on both sections of the transistor types covered herein, except where the electrical characteristic being evaluated applies to the transistor as a device entity.

4.5.3 Burn-in and operating life-test conditions. Power shall be applied to the device to achieve a junction temperature ($T_J$) of +187.5°C ±12.5°C.

$T_J$ shall be determined as follows:

$T_J = P_D R_{\theta JC} + T_C$
$P_D = (V_{CE})(I_C)$
$R_{\theta JC}$ = +0.65°C/W both sections
$T_C$ = measured value at hottest point on case

NOTE: For reference only. This test can be approximated by dissipating 4 watts at $T_A$ = +25°C.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage, collector to base | 3001 | Bias condition D<br>$I_C$ = 50 mA dc | $V_{(BR)CBO}$ | 60 | --- | V dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D<br>$I_C$ = 50 mA dc<br>Pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 30 | --- | V dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition C<br>$I_C$ = 50 mA dc | $V_{(BR)CES}$ | 60 | --- | V dc |
| Collector to base cutoff current | 3036 | Bias condition D<br>$V_{CB}$ = 30 V dc, $I_E$ = 0 | $I_{CBO}$ | --- | 1.5 | mA |
| Emitter to base cutoff current | 3061 | Bias condition D<br>$V_{EB}$ = 4 V dc | $I_{EBO}$ | --- | 5 | mA |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1 A dc<br>$I_B$ = 0.1 A dc<br>Pulsed (see 4.5.1)<br>(each section) | $V_{CE(sat)}$ | --- | 2.2 | V |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc<br>$I_C$ = 1 A dc,<br>Pulsed (see 4.5.1) | $h_{FE}$ | 20 | 100 | --- |
| Forward-current transfer ratio (gain ratio) | 3076 | $V_{CE}$ = 5 V dc<br>$I_C$ = 1 A dc<br>Pulsed (see 4.5.1) | $\frac{h_{FE1}}{h_{FE2}}$ | --- | 1.35 | --- |
| Power output test | | | | | | |
| 2N6985 | | | | | | |
| Electrical ruggedness | | $V_{CC}$ = 28 V dc<br>f = 400 MHz<br>$P_{out}$ = 125 W<br>Test circuit (see figure 2) 2/ | $\lambda$ | --- | --- | --- |
| Output power | | $V_{CC}$ = 28 V dc<br>f = 400 MHz<br>$P_{out}$ = 125 W<br>Test circuit (see figure 2) | $G_{PE}$ | 8.0 | --- | dB |
| Collector efficiency | | $V_{CC}$ = 28 V dc<br>f = 400 MHz<br>$P_{out}$ = 125 W<br>Test circuit (see figure 2) | $\lambda$ | 50 | --- | % |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| 2N6986 | | | | | | |
| Electrical ruggedness | | $V_{CC}$ = 28 V dc<br>f = 500 MHz<br>$P_{out}$ = 100 W<br>Test circuit (see figure 3) 2/ | λ | --- | --- | --- |
| Output power | | $V_{CC}$ = 28 V dc<br>f = 500 MHz<br>$P_{out}$ = 100 W<br>Test circuit (see figure 3) | $G_{PE}$ | 7.5 | --- | dB |
| Collector efficiency | | $V_{CC}$ = 28 V dc<br>f = 500 MHz<br>$P_{out}$ = 100 W<br>Test circuit (see figure 3) | λ | 50 | --- | % |
| Subgroup 3 | | | | | | |
| High-temperation operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D<br>$V_{CB}$ = 30 V dc | $I_{CBO}$ | --- | 50 | mA |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc<br>$I_C$ = 1.0 A dc<br>Pulsed (4.5.1) | $h_{FE}$ | 10 | --- | --- |
| Subgroup 4 | | | | | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 28 V dc<br>$I_E$ = 0<br>f = 1 MHz<br>(each section) | $C_{obo}$ | | 115 | pF |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.
2/ Devices circuit is switched to a load having a VSWR 30:1 and driven through all phase angles in a time interval not less than 3 seconds case temperature max 50°C. End-point test: Devices must meet power gain and collector efficiency test as specified above. This test also replaces the thermal response test.



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TABLE IIa. Group B inspection for JANS devices.

| Inspection 1/ | MIL-STD-750 | |
|---|---|---|
| | Method | Conditions |
| Subgroup 1 | | |
| Physical dimensions | 2066 | |
| Subgroup 2 | | |
| Solderability | 2026 | |
| Resistance to solvents | 1022 | |
| Subgroup 3 | | |
| Thermal shock (temperature cycling) | 1051 | |
| Hermetic seal<br>a. Fine leak<br>b. Gross leak | 1071 | |
| Electrical measurements | | See table V, steps 1, 2, 3, and 4 |
| Decap internal visual (design verification) | 2075 | 1 device /0 failures each lot |
| Bond strength | 2037 | Test condition A; all internal leads for each device shall be pulled separately |
| Die shear (excluding axial leaded services) | 2017 | Only devices previously subjected to the bond strength test shall be used for this test. |
| Subgroup 4 | | |
| Not applicable | | |
| Subgroup 5 | | |
| Accelerated steady-state operation life | 1027 | $V_{CB} \geq$ 10 V dc<br>$R_{\theta JC}$ = +0.65°C/W (both sections)<br>$T_J$ = 187.5°C ±12.5°C (see 4.5.3)<br>Marking legibility requirements shall not apply. |
| Electrical measurements | | See table V, all steps |
| Bond strenght (Al-Au die interconnects only) | | |
| Subgroup 6 | | |
| Not applicable | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE IIb. Group B inspection for JANTX and JANTXV devices.

| Inspection 1/ | MIL-STD-750 | |
|---|---|---|
| | Method | Conditions |
| Subgroup 1 2/ | | |
| Solderability | 2026 | |
| Resistance to solvents | 1022 | |
| Subgroup 2 | | |
| Thermal shock (temperature cycling) | 1051 | |
| Hermetic seal<br>a. Fine leak<br>b. Gross leak | 1071 | |
| Electrical measurements | | See table V, steps 1, 2, 3, and 4 |
| Subgroup 3 | | |
| Steady-state operation life | 1027 | $V_{CB} \geq 10$ V dc, $R_{\theta JC}$ = +0.65°C/W both sections, $T_J$ = 187.5°C ±12.5°C (see 4.5.3) |
| Electrical measurements | | See table V, all steps |
| Subgroup 4 | | |
| Decap internal visual (design verification) | 2075 | |
| Bond strength (wire or clip bonded) | 2037 | |
| Subgroup 5 | | |
| Not applicable | | |
| Subgroup 6 | | |
| High-temperature life (nonoperating) | 1032 | $T_{stg}$ = +200°C<br>t = 340 hours |
| Electrical measurements | | See table V, all steps |

1/ For sampling plan, see MIL-S-19500.
2/ Separate samples may be used for each test.



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TABLE III. Group C inspection (all quality levels).

| Inspection 1/ | MIL-STD-750 | |
|---|---|---|
| | Method | Conditions |
| Subgroup 1 | | |
| Physical dimensions | 2066 | |
| Subgroup 2 | | |
| Thermal shock (glass strain) | | Not applicable |
| Terminal strength | 2036 | Test condition A<br>W = 8 ±0.5 lbs<br>t = 5 sec. min. |
| Hermetic seal<br>a. Fine leak<br>b. Gross leak | 1071 | |
| Moisture resistance | 1021 | |
| Electrical measurements | | See table V, steps 1, 2, 3, and 4 |
| Subgroup 3 | | |
| Shock | 2016 | |
| Vibration, variable frequency | 2056 | |
| Constant acceleration | 2006 | 10,000 G's |
| Electrical measurements | | See table V, steps 1, 2, 3, and 4 |
| Subgroup 4 | | |
| Salt atmosphere (corrosion) | 1041 | |
| Subgroup 5 | | |
| Not applicable | | |
| Subgroup 6 | | |
| Steady-state operation life | 1027 | $V_{CB} \geq$ 10 V dc<br>$R_{\theta JC}$ = +0.65°C/W (both sections)<br>$T_J$ = 187.5°C<br>12.5°C (see 4.5.3) |
| Electrical measurements | | See table V, steps 1, 2, 3, 4, 5, 6, and 8 |

1/ For sampling plan, see MIL-S-19500.



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TABLE IV. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Qualification and large lot quality conformance inspection | Small lot quality conformance inspection n/c |
|---|---|---|---|---|
| | Method | Conditions | | |
| Subgroup 1 | | | 10 devices, c = 0 | 4 devices, c = 0 |
| Thermal shock (glass strain) | 1056 | 0°C to 100°C, 100 cycles | | |
| Electrical measurements | | See table V, steps 1, 2, and 3 | | |
| Subgroup 2 | | | 10 devices, c = 0 | 4 devices, c = 0 |
| Steady-state dc blocking life | 1039 or 1049 | Condition A; 1000 hours | | |
| Electrical measurements | | See table V, steps 1 and 4 | | |
| Subgroup 3 | | | 2 devices, c = 0 | 1 device, c = 0 |
| DPA | 2102 | | | |
| Subgroup 4 | | | 6 devices, c = 0 | 3 devices, c = 0 |
| Thermal resistance | | $R_{\theta JC}$ = .65°C/W maximum per device ($I_R$ scan will be used to verify $R_{\theta JC}$) | | |
| Forward voltage drop (emitter to base diode method) | | See 4.3.2 | | |
| Subgroup 5 | | | | |
| Not applicable | | | | |



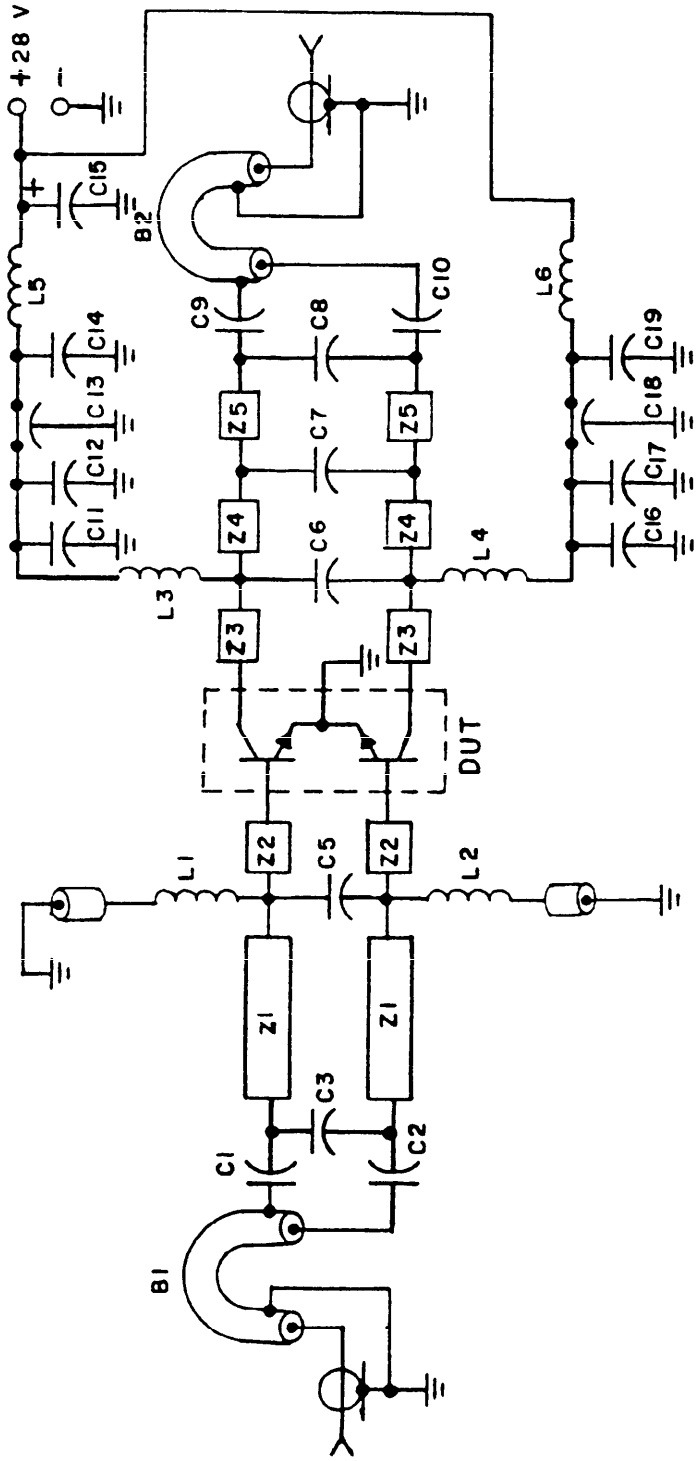
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TABLE V. Groups A, B, and C electrical measurements.

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1 | Breakdown to voltage collector to emitter | 3011 | Bias condition D, $I_C$ = 50 mA dc Pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 30 | --- | V dc |
| 2 | Collector cutoff current | 3036 | Bias condition D $V_{CB}$ = 30 V dc $I_E$ = 0 | $I_{CBO}$ | --- | 1.5 | mA dc |
| 3 | Emitter to base cutoff current | 3061 | Bias condition D $V_{EB}$ = 4 V dc | $I_{EBO}$ | --- | 5 | mA dc |
| 4 | Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc $I_C$ = 1 A dc Pulsed (see 4.5.1) | $h_{FE}$ | 20 | 100 | --- |
| 5 | Collector cutoff current | 3036 | Bias condition D $V_{CB}$ = 30 V dc $I_E$ = 0 | $\Delta I_{CBO}$ | 100% of initial value of 150 µA, which is greater. | | |
| 6 | Foward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc $I_C$ = 1 A dc Pulsed (see 4.5.1) | $\Delta h_{FE}$ | 25% subgroup 2 of table I herein. | | |
| 7 | Output power | | $V_{CC}$ = 28 V dc | | | | |
| | 2N6895 | | f = 400 MHz $P_{out}$ = 125 W Test circuit (see figure 2) | $G_{PE}$ | 8.0 | --- | dB |
| | 2N6896 | | f = 500 MHz $P_{out}$ = 100 W Test circuit (see figure 3) | $G_{PE}$ | 7.5 | --- | dB |
| 8 | Collector efficiency | | $V_{CC}$ = 28 V dc | | | | |
| | 2N6985 | | f = 400 MHz $P_{out}$ = 125 W Test circuit (see figure 2) | $\lambda$ | 50 | --- | % |
| | 2N6986 | | f = 500 MHz $P_{out}$ = 100 W Test circuit (see figure 3) | $\lambda$ | 50 | --- | % |



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C1, C2 - 240 pF, 100 mil chip cap (ATC) or equivalent
C3 - 4.7 pF, 100 mil chip cap (ATC) or equivalent
C8 - 12 pF, 100 mil chip cap (ATC) or equivalent
C5 - 27 pF, 100 mil chip cap (ATC) or equivalent
C6 - 20 pF, 100 mil chip cap (ATC) or equivalent
C7 - 12 pF, 100 mil chip cap (ATC) or equivalent
C9, C10 - 270 pF, 100 mil chip cap (ATC) or equivalent
C11, C12, C16, C17 - 470 pF 100 mil chip cap (ATC) or equivalent
C13, C18 - 680 pF feedthru
C14, C19 - 0.1 µF Erie Redcap or equivalent
C15 - 20 µF, 50 V

B1 - balun, 50 Ω semi-rigid coaxial cable 86 mil OD, 2 inches L
B2 - balun, 50 Ω semi-rigid coaxial cable 86 mil OD, 2 inches L

L1, L2 - 0.15 µH molded choke with ferrite bead
L3, L4 - 2-1/2 turns no. 20 AWG .200 inch ID
L5, L6 - 3-1/2 turns no. 18 AWG .200 inch ID

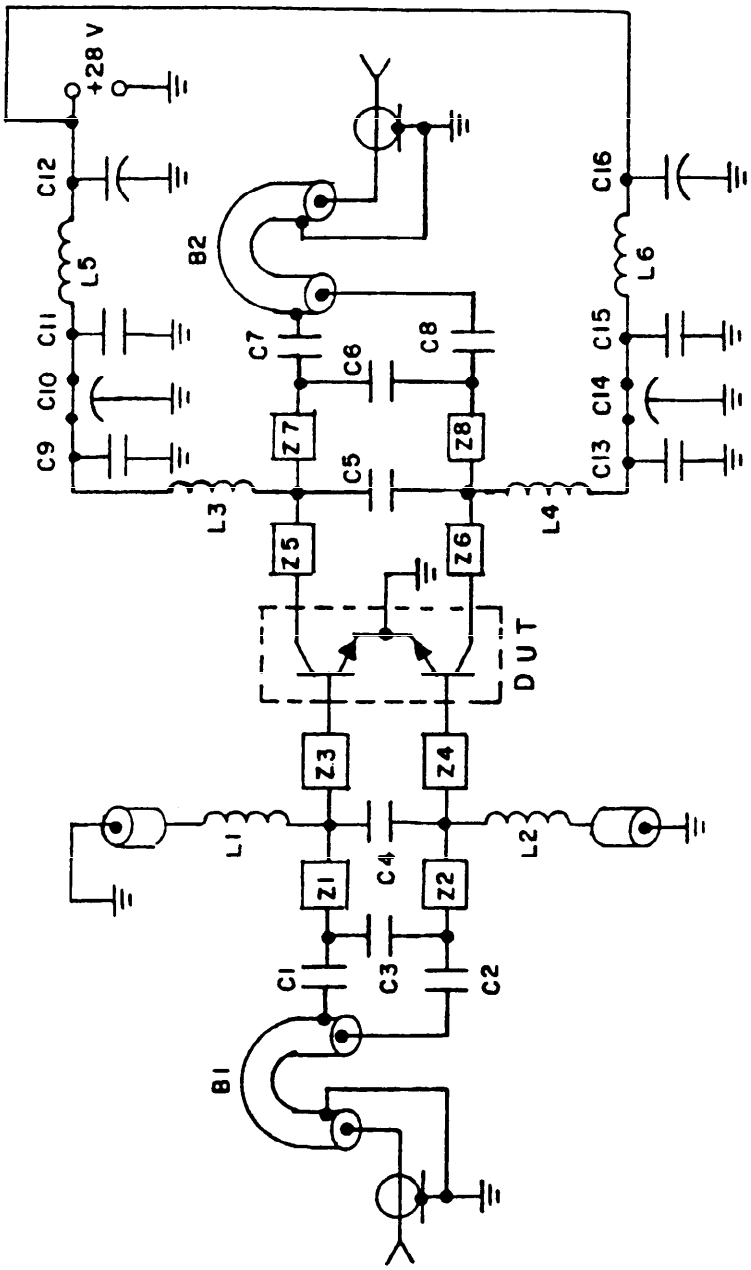
Z1 - microstrip line 650 mil L x 125 mil W
Z2 - microstrip line 220 mil L x 125 mil W
Z3 - microstrip line 280 mil L x 125 mil W
Z4 - microstrip line 300 mil L x 125 mil W
Z5 - microstrip line 450 mil L x 125 mil W

Board material - .0625 inch teflon-fiberglass,
$\varepsilon_r = 2.5 \pm .05$ 1 oz. copper clad,
both sides.

FIGURE 2. Test fixture (400 MHz).



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C1, C2, C7, C8 - 240 pF 100 mil chip cap
C3 - 12 pF 100 mil chip cap
C4 - 10 pF 100 mil chip cap
C5 - 36 pF 100 mil chip cap
C6 - 12 pF 100 mil chip cap
C9, C13 - 1000 pF 100 mil chip cap
C10, C14 - 680 pF feedthru cap
C11, C15 - 0.1 µF ceramic disc cap
C12, C16 - 50 µF 50 V

L1-L2 - 0.15 µH molded choke with ferrite bead
L3-L4 - 2-1/2 turns no. 20 AWG 0.200 inch ID
L5-L6 - 3-1/2 turns no. 18 AWG 0.200 inch ID

B1, B2 - balun 50 Ω semi rigid coax, 86 mil OD, 2 inches

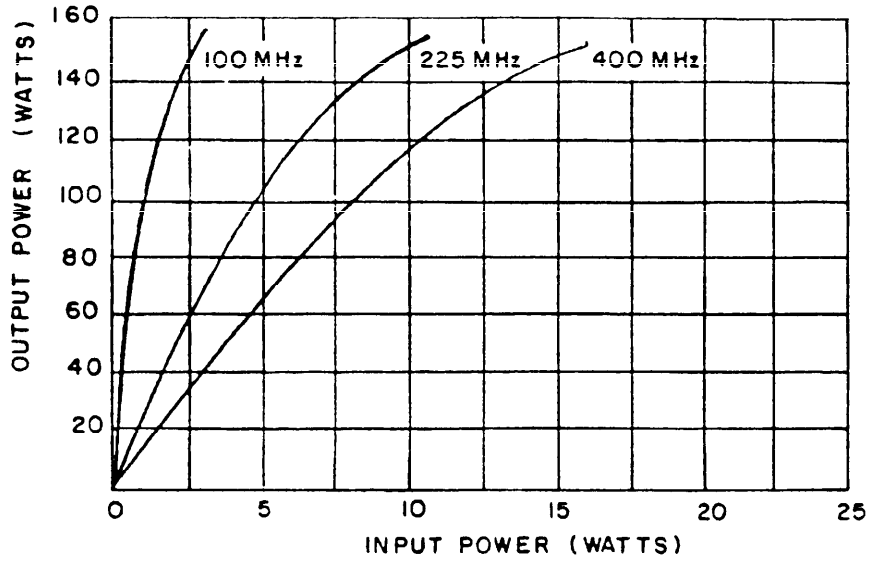
Z1, Z2 - 450 mil x 125 mil W microstrip
Z3, Z4 - 340 mil x 125 mil W microstrip
Z5, Z6 - 280 mil x 125 mil W microstrip
Z7, Z8 - 600 mil x 125 mil W microstrip

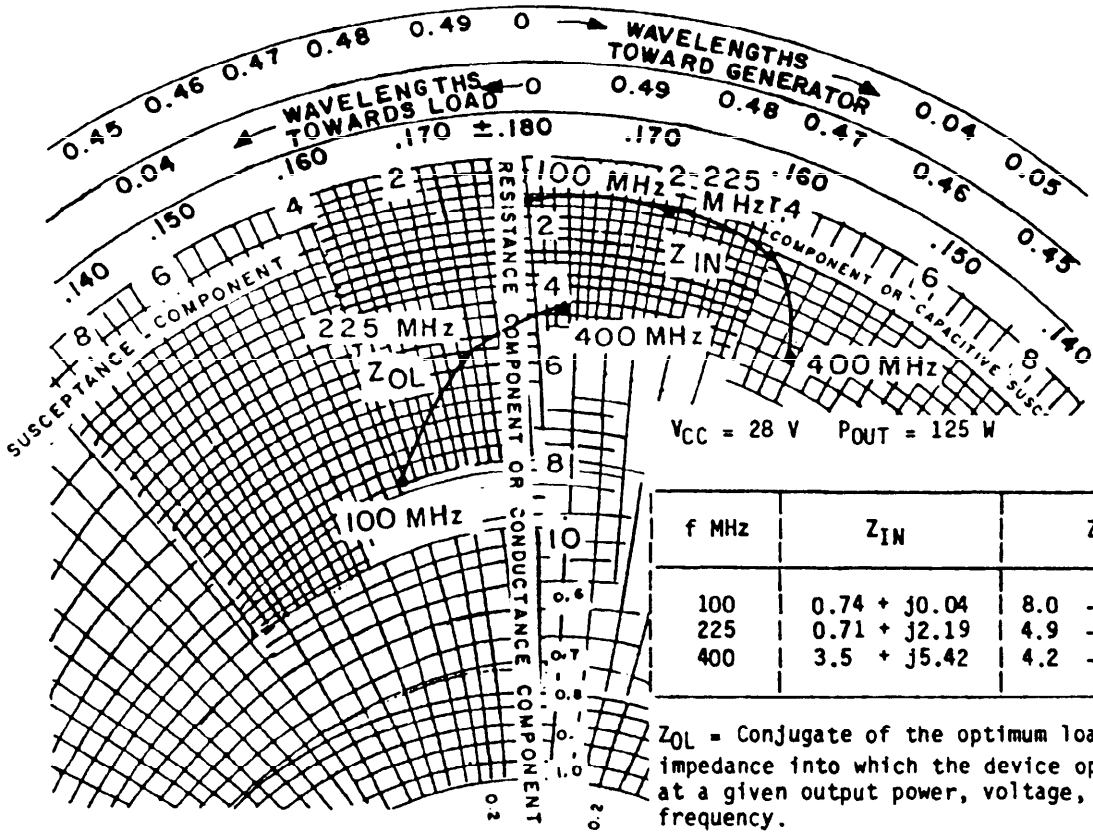
Board material - .0325 inch teflon-fiberglass ε = 2.56, 1 oz. copper clad both sides.

FIGURE 3. Test fixture (500 MHz).



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| f MHz | $Z_{IN}$ | $Z_{OL}$ |
|---|---|---|
| 100 | 0.74 + j0.04 | 8.0 - j3.9 |
| 225 | 0.71 + j2.19 | 4.9 - j0.88 |
| 400 | 3.5 + j5.42 | 4.2 - j0.93 |

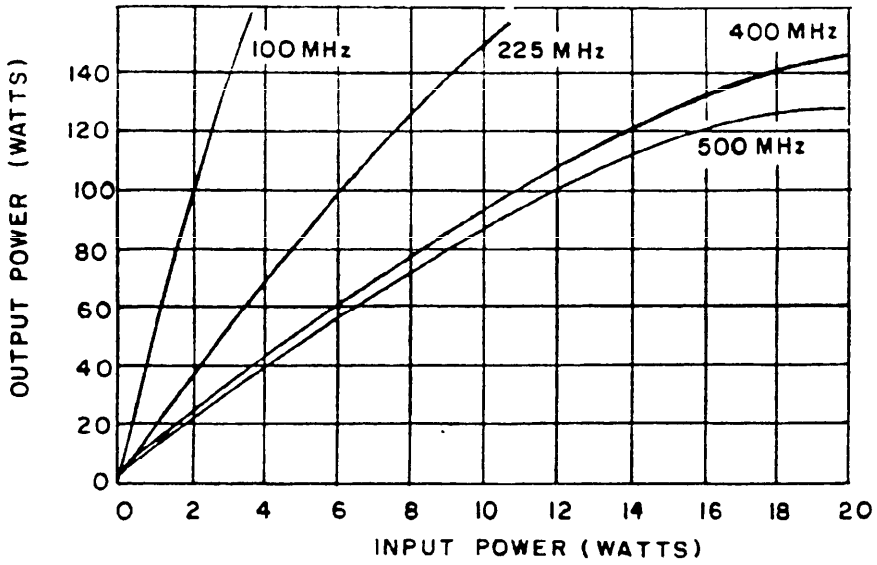
$Z_{OL}$ = Conjugate of the optimum load impedance into which the device operates at a given output power, voltage, and frequency.

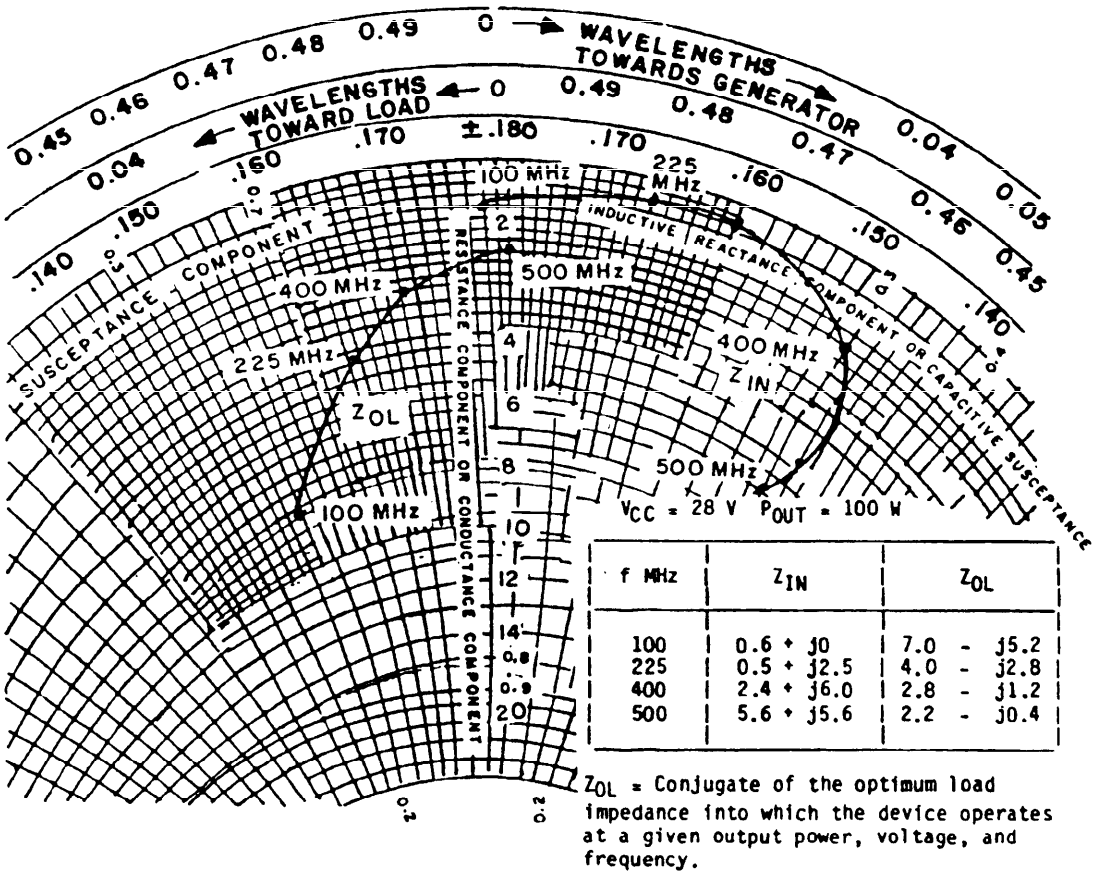
NOTE: Input and output impedances are measured from base to base and collector to collector respectively.

FIGURE 4. 2N6985.



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| f MHz | $Z_{IN}$ | $Z_{OL}$ |
|---|---|---|
| 100 | 0.6 + j0 | 7.0 - j5.2 |
| 225 | 0.5 + j2.5 | 4.0 - j2.8 |
| 400 | 2.4 + j6.0 | 2.8 - j1.2 |
| 500 | 5.6 + j5.6 | 2.2 - j0.4 |

$Z_{OL}$ = Conjugate of the optimum load impedance into which the device operates at a given output power, voltage, and frequency.

NOTE: Input and output impedances are measured from base to base and collector to collector respectively.

FIGURE 5. 2N6986.



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Typical power output and impedance curves. Typical power output and impedance curves for the 2N6985 and 2N6986 are shown are information only, see figures 4 and 5.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part number. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part number.

Supersession and cross reference data

| Military part number | Manufacturer's CAGE code | Manufacturer's and user's part number |
|---|---|---|
| | | |



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1139)



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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/584 | 2. DOCUMENT DATE *(YYMMDD)* |
|---|---|---|

3. DOCUMENT TITLE
Semiconductor Device, RF Power Transistor, NPN, Silicon, Push-Pull

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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